

Lab Sheet-4

Aim: Implementation of User Creation, Deletion, and Privilege Management in SQL

Problem Statement:

Design and execute SQL commands to create users, manage their access privileges, and remove users in a database system. This assignment focuses on database security and access control mechanisms.

Tasks

Part A: User Creation

1. Create a new user named user1 with a password.
2. Create another user user2 with limited access.

Code:

```
1 • CREATE USER 'user1'@'localhost' IDENTIFIED BY '1234';  
2 • CREATE USER 'user2'@'localhost' IDENTIFIED BY '1234';
```

Part B: Granting Privileges

3. Grant SELECT and INSERT privileges on a table Student to user1.
4. Grant ALL privileges on a table Course to user2.
5. Provide WITH GRANT OPTION to user1 and observe its effect.

```

1 • GRANT SELECT, INSERT
2   ON Student
3   TO 'user1'@'localhost';
4
5 • GRANT ALL PRIVILEGES
6   ON Course
7   TO 'user2'@'localhost';
8
9 • GRANT SELECT, INSERT
10  ON Student
11  TO 'user1'@'localhost'
12  WITH GRANT OPTION;

```

Part C: Revoking Privileges

6. Revoke INSERT permission from user1 on the Student table.

7. Revoke all privileges from user2 on the Course table.

```

1 • REVOKE INSERT
2   ON Student
3   FROM 'user1'@'localhost';

```

```

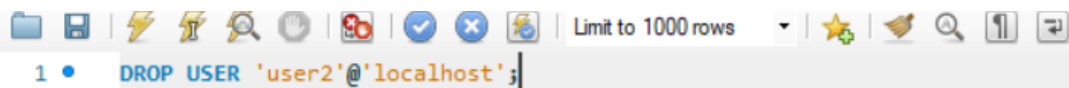
1 • REVOKE ALL PRIVILEGES
2   ON Course
3   FROM 'user2'@'localhost';

```

Part D: User Deletion

8. Drop/Delete user user2 from the database.

9. Attempt to login using deleted credentials and observe the result.



```

1 • DROP USER 'user2'@'localhost';

```

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

```
1 SELECT user, host
2 FROM mysql.user;
3 • SHOW GRANTS FOR 'user1'@'localhost';
```

Result Grid	Filter Rows:	Export: 	Wrap Cell Content: 	 Result Grid
Grants for user1@localhost				
▶	GRANT USAGE ON *,* TO `user1`@`localhost`			
	GRANT SELECT ON `college_db`.`student` TO ...			